

Impression Confusion: Soft Liners as Impression Materials

Group 8

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Faculty Mentor: Dr. Griffin



5 Behavioral Objectives

- **Describe** the biological and clinical principles of functional impression techniques in edentulous patients, including the use of soft denture liners as long-term impression, **functional** impression materials.
- **Differentiate** between commercially available soft liner products by explaining their composition, intended use, and off-label application for functional impressions.
- **Evaluate** the potential health risks associated with soft liner materials, particularly the presence of phthalates and other plasticizers, and their implications for patient safety.
- **Compare and contrast** the effectiveness and limitations of soft liners versus alternative impression materials and techniques based on current scientific evidence and product availability.
- **Apply** a critical decision tree to **select** appropriate functional impression approaches for edentulous patients, balancing clinical outcomes, material properties, safety concerns, and regulatory considerations.

BO1. Describe the biological and clinical principles of functional impression techniques in edentulous patients, including the use of soft denture liners as long-term impression materials.



Search Strategy



P: Problem (What is the issue?)

Ill denture fit

I: Intervention (Which material or technique is being tested?)

Impression materials

C: Comparison (What is the alternative material or technique?)

Soft liners vs. traditional methods

O: Outcome (What is the desired result or effect?)

Improved denture fit

Pubmed search strings

(functional) AND (dental impression)
AND (materials) AND (complete
denture)- 138 results

(soft denture liner)- 5yrs- 71 results

(soft denture liner) AND
(impression)- 29 results

Databases used

Pubmed

Journal of Prosthetic Dentistry

Highest level of evidence

Systematic reviews within 5 years



What makes an impression *truly* good?



Retention

Stability

Support

Esthetics

Preservation of remaining structures



Brief History of Edentulous Impressions

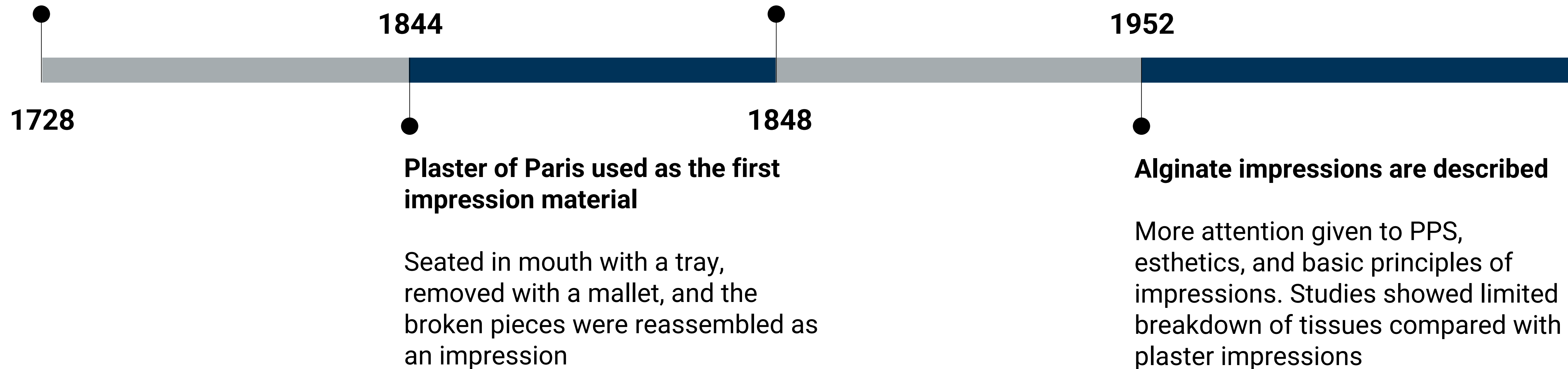


Pierre Fauchard

Measured the mouth with a compass and carved an approximate shape into bone or ivory. Held into place with wire and teeth were made of ext human or animal teeth

Gutta Percha/Beeswax/Modeling Compound introduced as an impression material

Placed in boiling water and kneaded until it was inserted in the mouth. Often had a final wash of plaster



Principles/Theories of Impression Making



Mucostatic

Adapting the impression material to the tissues to capture them in their most undistorted form

Mucocompressive

All tissues recorded evenly under equal pressure despite their anatomy and role in retention

Selective pressure

Records stress bearing areas under pressure and non-stress bearing areas under minimum pressure



Functional Impression Techniques



The term “**functional impression**” can be used to describe **active (patient)** or **passive (operator)** functional movements, or both simultaneously.

FUNCTIONAL RELINE

Relining the old denture with a tissue conditioner or soft liner and dismissing the patient for 2 weeks

FLUID WAX

Initial wash with zinc eugenol and final touches with a fluid wax (Korecta no.4 or Iowa wax) to capture dynamic movements of movable structures

CLOSED MOUTH TECHNIQUE

Initial RBOR fabricated and patient closes to engage and record functional movements under the RB

OPEN MOUTH TECHNIQUE (DCG)

Custom tray fabricated, tray seated with patient's mouth open, combination of active and passive movements



Functional Impression Techniques



HONORABLE MENTIONS:

TURBYFILL METHOD

WINDOW TECHNIQUE (for flabby ridge)

- Custom tray fabricated with a “window” or opening where the flabby ridge is exposed
- Initial wash completed with a low viscosity material and flabby ridge captured with high viscosity material to not displace flabby ridge

NEUTRAL ZONE TECHNIQUE (for atrophic ridges)

- NZ defined as the space in the oral cavity where the outward forces from the tongue and inward forces of cheeks/lips are nullified during functional movements
- Involves making a “dough-like” material using a 3:7 ratio of impression compound to modeling compound
- patient instructed to do a number of functional movements (sucking, swallowing, EEE, OOO) to capture NZ



What are denture liners?



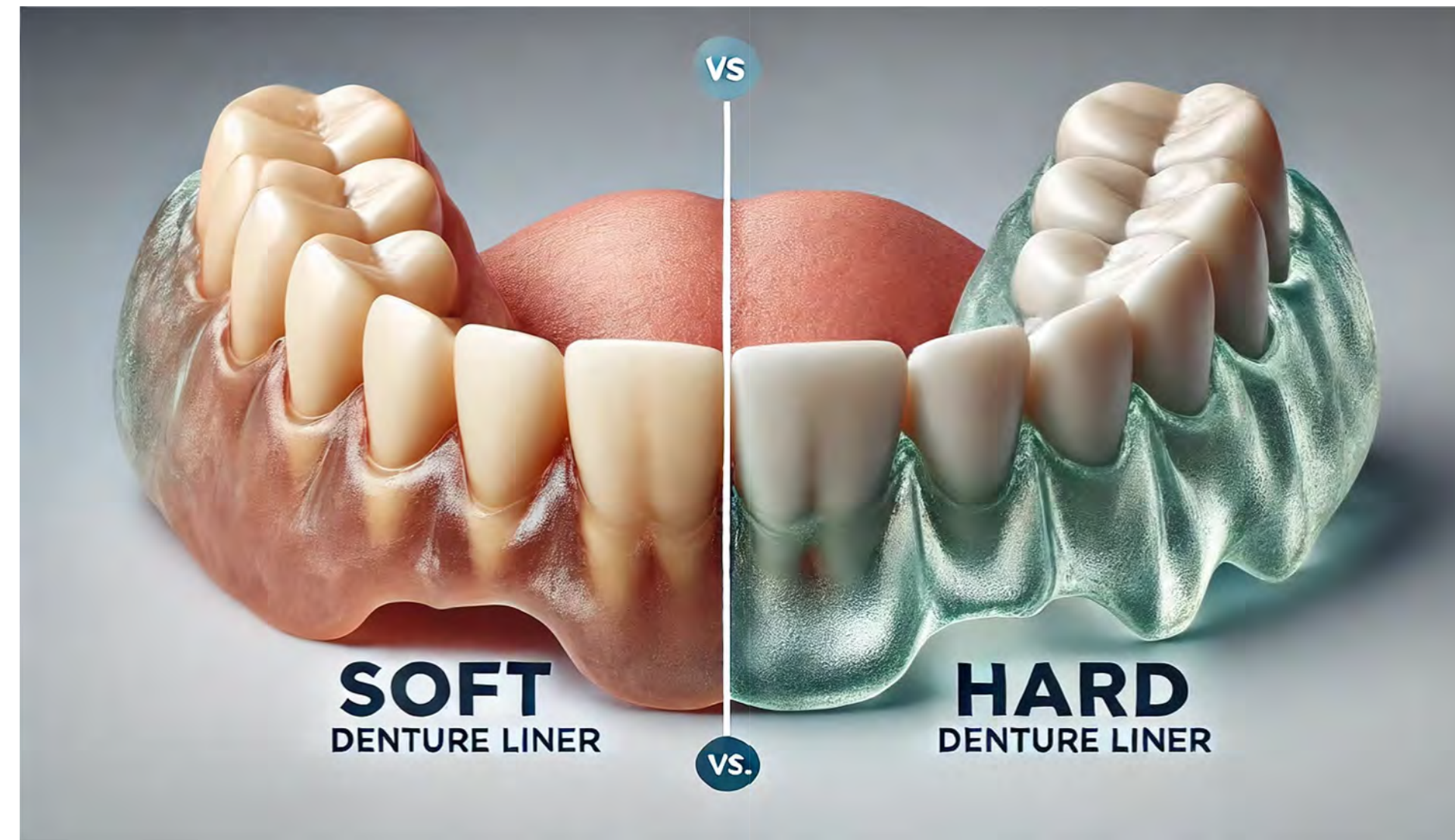
- Edentulous patients will experience ridge resorption and wear over time, causing the fit of their complete dentures to change.
- Can remake the denture completely, but in cases of intact denture teeth and maintained esthetics, relining can be more affordable.
 - Complete dentures can cost thousands of dollars, denture relining runs for a few hundred dollars
- CHOICES: both soft and hard are both soft denture liners exist. Both add a layer of material to the intaglio surface of the denture to improve comfort, fit, and therefore usability of the patient's existing denture.
 - DENTURE CONDITIONERS are very short-term and palliative soft liner, only meant to heal irritated tissues prior to final impressions



[simply-natural-digital-dentures-2560x2025.png](#)



Types of Liners



Soft Liners:

- medical silicone or plasticized acrylics
 - cushioning, flexible, need replacing
- for immediate or short-term prostheses

Hard Liners:

- PMMA or polymethacrylate resin
- rigid, stable, long-lasting, cleansable
- can last many years, more permanent revision of long-term prostheses

Combo?

BO2. Differentiate between commercially available soft liner products by explaining their composition, intended use, and off-label application for functional impressions.



Search Strategy

P: Patient/population

- edentulous patients

I: Intervention

- Soft denture liners

C: Comparison

- Different categories of soft liners (ex: silicone-based, acrylic-based)

O: Outcome

- Composition, intended clinical use, and off-label application for functional impressions

Databases used

Pubmed

Galileo

Parameters

10 years

Article types →

- ☒ Books and Documents
- ☒ Clinical Trial
- ☒ Meta-Analysis
- ☒ Randomized Controlled Trial
- ☒ Review
- ☒ Systematic Review

Pubmed search string

Parameters: 10 years,

Denture soft liner → **15 results**

Soft denture liner AND acrylic → **6 results**

Soft denture liner AND hardness → **2 results**

Highest level of evidence

Systematic reviews, randomized clinical trials



Intended Use of Soft liners



Used in complete and removable partial dentures to evenly distribute pressure to all denture bearing surfaces in the oral cavity.

- Form a soft cushion between the denture base and oral mucosa

When are soft liners used?

- “When the patient cannot tolerate the stress induced by dentures.” (Dr. Clary)
- “Irregular bone resorption, bony undercuts, thin atrophic mucosa, immediate prosthesis, healing after implant placement, and for patients with bruxism and xerostomia.” (Hashem 2025)

Classification of Soft Liners



Classification types: material, processing method, length of use

Material types:

Acrylic

Powder/liquid systems

- Powder = polymer
- Liquid = monomer + plasticizer

Silicone

Polydimethylsiloxane polymers

Alkyl Silicate Cross linking agent

Processing method:

Auto Polymerized - “chairside”

Heat cured - “lab processed reline”

Classification of Soft Liners Continued

Length of Use: Short Term

- Also called “tissue conditioners”
- Usually used right after surgery
 - Do not contain methacrylate monomers → prevents cytotoxicity and irritation to healing tissues.
 - Contain poly ethyl methacrylate (PEMA) powder, aromatic esters, and alcohol → no polymerization reaction
- Used for a few days and up to one week

Long Term

- Acrylic or silicone
 - Acrylic → used for up to 6 months
 - Silicone → used for up to 1 year



Tissue conditioner



Long term soft liner



What Do We Want In A Denture Liner?

Ideal Viscoelasticity: flow readily under pressure, dimensional stability (weight change, water sorption, solubility), minimal elastic recovery, close adaptation to tissues

Antimicrobial

Rehabilitative: Patient satisfaction/comfort to ease soreness/irritation from ill-fitting dentures

Resilient: Energy absorption for ridge protection, evenly distributing stresses/ selective pressure

High Tensile Bond Strength to Denture Base

Hardness: depends on the situation (comfort (low) or longevity (high))

Efficient Timing

Acrylic vs Silicone Soft Liners



Systematic Review originally identifying 176 studies that was narrowed down to 36 studies comparing acrylic and silicone soft reliners

Silicone

- Lower water sorption
- Higher tensile & shear bond strength
- More color stable, resistant to staining
- Lower surface roughness & hardness
- More resistant to cleanser degradation
- Elastic behavior → more durable over time

Acrylic

- Viscoelastic behavior → easier short-term use
- Lose softness and elasticity over time
- Plasticizer leaching leads to hardening

Acrylic vs Silicone Soft Liners

- Double-blind, randomized clinical trial comparing acrylic-based and silicone-based soft liners
- **Oral health-related quality of life**
 - Both acrylic-based and silicone-based soft liners significantly improved OHRQoL (higher scores indicate worse OHRQoL)
- **Maximum bite force**
 - Both significantly improved OHRQoL
 - At 3 months, the silicone-based soft liners showed significantly greater bite force
- Silicone performed better than acrylic over time because:
 - Lower water absorption/solubility
 - Better stability of elastic/viscoelastic properties
 - Acrylic loses plasticizer, absorbs water, and loses viscoelastic properties
- MD = mean difference between the 2 groups
- (CI) = confidence interval --> range of values that the MD could realistically be



Table 2. The overall oral health-related quality of life (Mean ± SD) at baseline and one and three months after denture relines.

Interventions	Baseline	One Month	Three Months
Acrylic-Based SL	56.3 ± 8.2 Aa	34.6 ± 7.8 Bb	29.4 ± 5.9 Bb
Silicone-Based SL	53.1 ± 12.8 Aa	31.9 ± 7.4 Bb	25.8 ± 6.2 Bb
MD (CI)	-3.2 (-11.5 to 5.1)	-2.7 (-8.6 to 3.2)	-3.6 (-8.3 to 1.1)

Table 4. Maximum bite force in Newtons (Mean ± SD) at baseline; 1, 2, and 3 months after denture adjustment.

Interventions	Baseline	One Month	Three Months
Acrylic-Based SL	75 ± 31 Aa	145 ± 53 Bb	116 ± 47 Bc
Silicone-Based SL	83 ± 32 Aa	156 ± 49 Bb	166 ± 57 Cb
MD (CI)	8 (-16.4 to 32.4)	11 (-28.6 to 50.6)	50 (9.4 to 60.5)

Comparing Bond Strength and Hardness

1. **Bond strength** → any favorable properties of a good liner are useless in the absence of a good bond to the denture base material

2. **Long term softness** → think about the purpose of the material

Materials used:

- Vertex soft: acrylic, heat
- Coe soft: acrylic, auto
- Molloplast B: silicone, heat
- Mollosil plus: silicone, auto

Results:

- increased hardness,
decreased bond strength

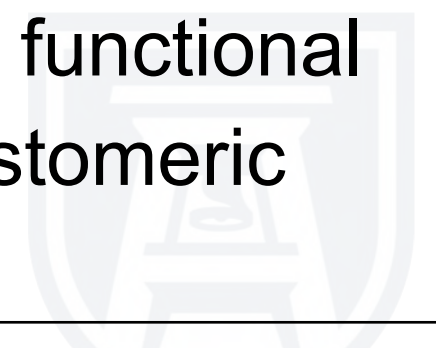
TABLE IV. Mean (SD) values of tensile bond strength (MPa) and hardness (Shore units) of resilient liner materials for 5 time intervals (n=10)

	Bond Strength				Hardness			
	Vertex Soft	Coe-Soft	Molloplast-B	Mollosil Plus	Vertex Soft	Coe-Soft	Molloplast-B	Mollosil Plus
1 day	3.50 ^{Ca} (0.44)	0.45 ^{Cb} (0.18)	1.58 ^{Cc} (0.19)	1.20 ^{Bd} (0.41)	49.08 ^{Aa} (1.34)	12.37 ^{Ab} (0.81)	42.28 ^{Ac} (0.89)	29.33 ^{Ad} (1.43)
1 week	3.07 ^{Ca} (0.46)	0.39 ^{Cb} (0.04)	1.42 ^{BCc} (0.31)	1.12 ^{Bd} (0.19)	49.38 ^{Aa} (1.15)	17.82 ^{Bb} (1.45)	42.42 ^{Ac} (1.46)	29.60 ^{ABd} (2.46)
1 month	2.57 ^{Ba} (0.29)	0.23 ^{Bb} (0.05)	1.16 ^{ABc} (0.12)	1.07 ^{Bd} (0.58)	51.22 ^{Aa} (2.98)	20.88 ^{Cb} (1.31)	43.02 ^{Ac} (1.44)	31.06 ^{Bd} (0.87)
3 months	1.91 ^{Aa} (0.71)	0.22 ^{Bb} (0.04)	1.10 ^{ABc} (0.53)	0.77 ^{Ad} (0.15)	50.88 ^{Aa} (4.48)	23.01 ^{Db} (1.43)	43.78 ^{Ac} (1.84)	33.94 ^{Cd} (1.26)
6 months	1.70 ^{Aa} (0.46)	0.11 ^{Ab} (0.01)	1.03 ^{Ac} (0.44)	0.50 ^{Ad} (0.22)	64.08 ^{Ba} (2.83)	24.02 ^{Db} (1.28)	46.16 ^{Bc} (2.15)	34.64 ^{Cd} (2.06)

Different superscripted uppercase letters indicate statistically different means within each column ($P < .001$).

Different superscripted lowercase letters indicate statistically different means within each row ($P < .001$).

<u>Product</u>	<u>Manufacturer</u>	<u>Adhesive / Bonding</u>	<u>Cure Type</u>	<u>Longevity</u>	<u>Intended Use</u>	<u>Functional Impression Use</u>
Molloplast-B	Detax	Silane coupling agent (Molloplast B primer or Primo Adhesive)	Heat-cure	Long (4–6 years)	Long-term permanent soft liner for chronic soreness, atrophic ridges	No official indication , but sometimes used academically for functional impressions due to stability (off-label)
Ufi Gel P	Voco	Ufi Gel SC adhesive	Auto	Medium (2 weeks–2 years)	Medium-term resilient liner Chairside relines	Not officially indicated but widely used in dental schools for functional impressions (off-label)
Flexibase	Flexico Developments Ltd.	Silane coupling agent	Auto	Long	Chairside long-term soft liner	Often used off-label for functional impressions due to elastomeric properties
Sofreliner Tough	Tokuyama	Primer	Auto	Long (~2 years)	Long-term chairside soft liner	Not officially indicated, but commonly used off-label for functional impressions
Luci-Sof	Dentsply Sirona	—	Heat	Long	Long-term permanent soft liner	No official indication for impressions



<u>Product</u>	<u>Manufacturer</u>	<u>Adhesive/Bonding</u>	<u>Cure Type</u>	<u>Longevity</u>	<u>Intended Use</u>	<u>Functional Impression Use</u>
Flexibase	(various/lab silicone)	Silane coupling agent	Auto	Long	Chairside long-term soft liner	Often used off-label for functional impressions due to elastomeric properties
Sofreliner Tough	Tokuyama	Primer	Auto	Long (~2 years)	Long-term chairside soft liner	Not officially indicated, but commonly used off-label
Luci-Sof	Dentsply Sirona	—	Heat	Long	Long-term soft liner	No official indication for impressions
Vertex Soft	Vertex Dental	MMA-based monomer bonding	Heat	Medium	Medium-term liner	Not indicated for functional impressions
Trusoft	Bosworth	Bonding liquid	Auto	Short (~30 days)	Tissue conditioner , short-term liner, pre-conditioning	Occasionally used as functional impression material , but not officially indicated
PermaSoft	(various)	Auto bond	Auto	Medium	Soft liner	Not indicated for functional impressions
Coe-Soft	GC	None (mechanical)	Auto	Short (~3 months)	Tissue conditioning , interim soft liner	Not indicated as the impression material
Coe-Comfort Visco-Gel FITT Lynal	GC	None (mechanical)	Auto	Very short (days–weeks)	Tissue conditioner , acutely inflamed tissues	YES→ some explicitly marketed for functional impressions (FITT and Visco-GEL marketed as tissue-conditioning functional impression material)



BO3. Evaluate the potential health risks associated with soft liner materials, particularly the presence of phthalates and other plasticizers, and their implications for patient safety.



Search Strategy

P: Problem (What is the issue?)

Phthalates/plasticizers in soft denture liners

I: Intervention (Which material or technique is being tested?)

Soft denture liners

C: Comparison (What is the alternative material or technique?)

Acrylic vs. silicone liners

O: Outcome (What is the desired result or effect?)

Reduced health risks

Databases used

Pubmed

Pubmed search strings

("Phthalates") OR ("plasticizer) AND
("Acrylic denture liners")

("Phthalates") OR ("plasticizer) AND
("soft denture liner")

("phthalates") AND ("toxicity")

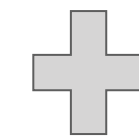
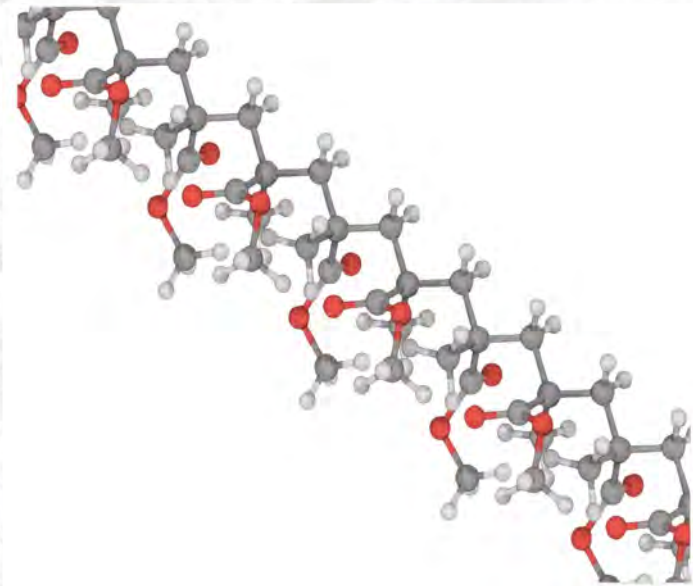
Highest level of evidence

Systematic reviews within 5 years
Meta-analyses

Acrylic Liners: Overview

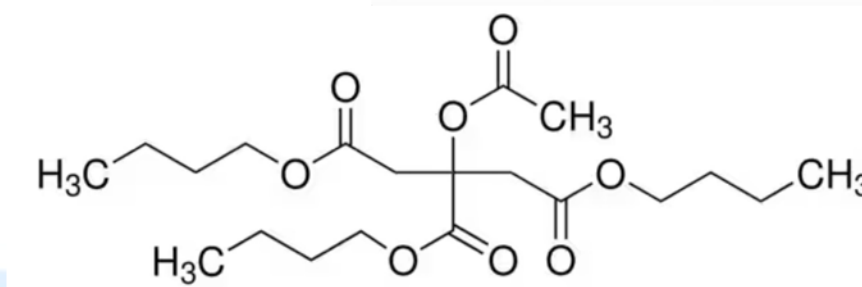
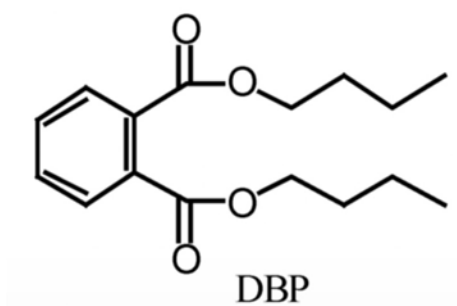
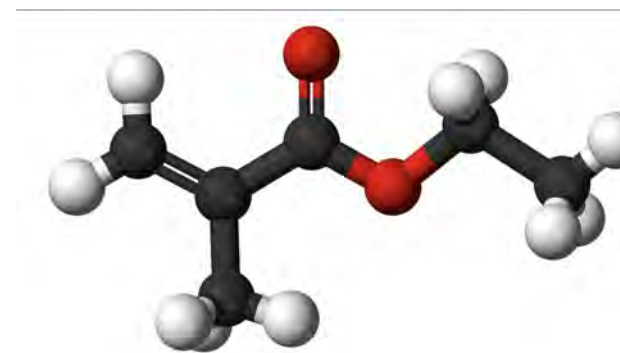
Powder

- Acrylic polymer: poly-ethyl methacrylate/poly-methyl methacrylate
- Radical initiator (ie organic peroxide)

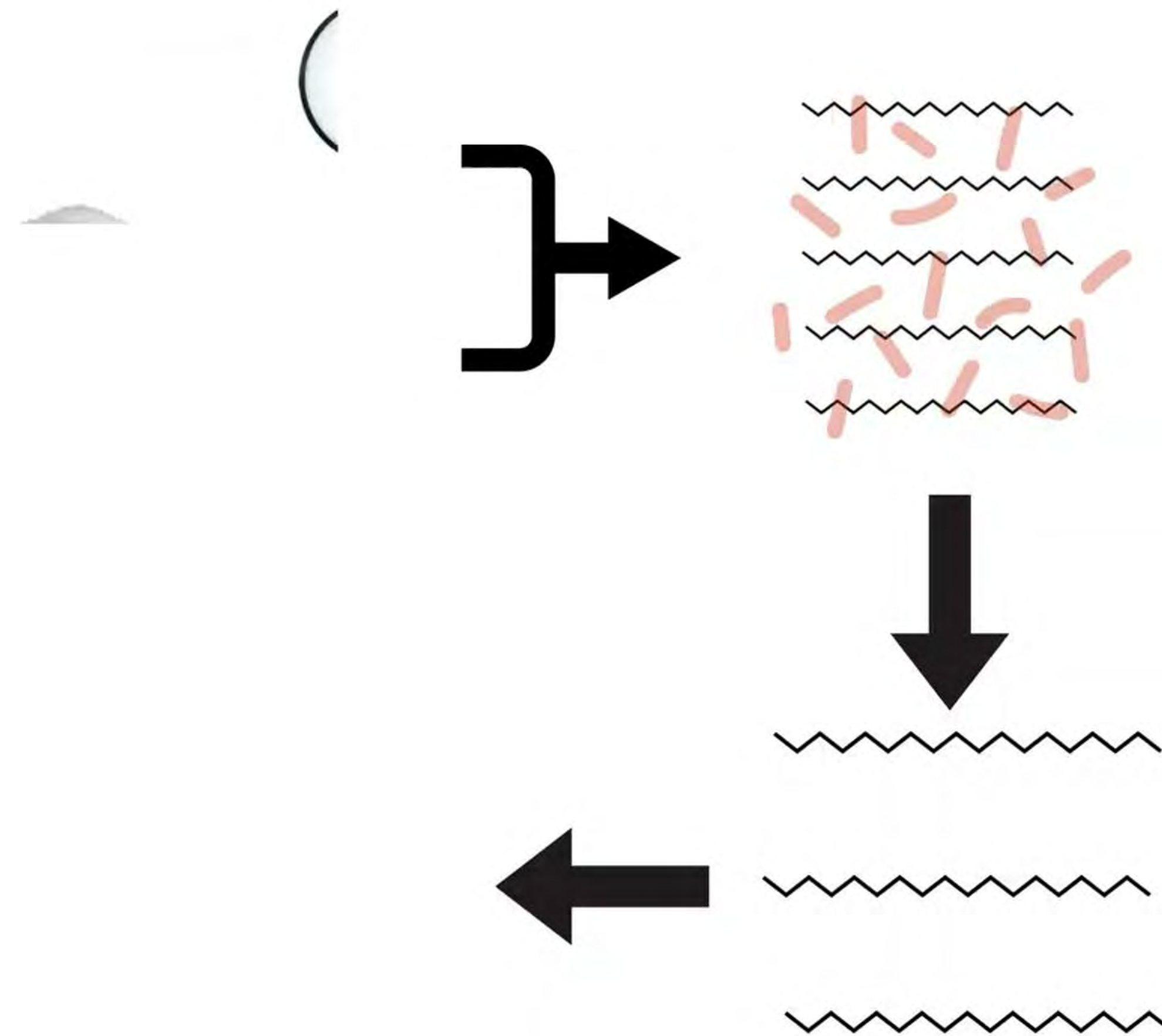


Liquid

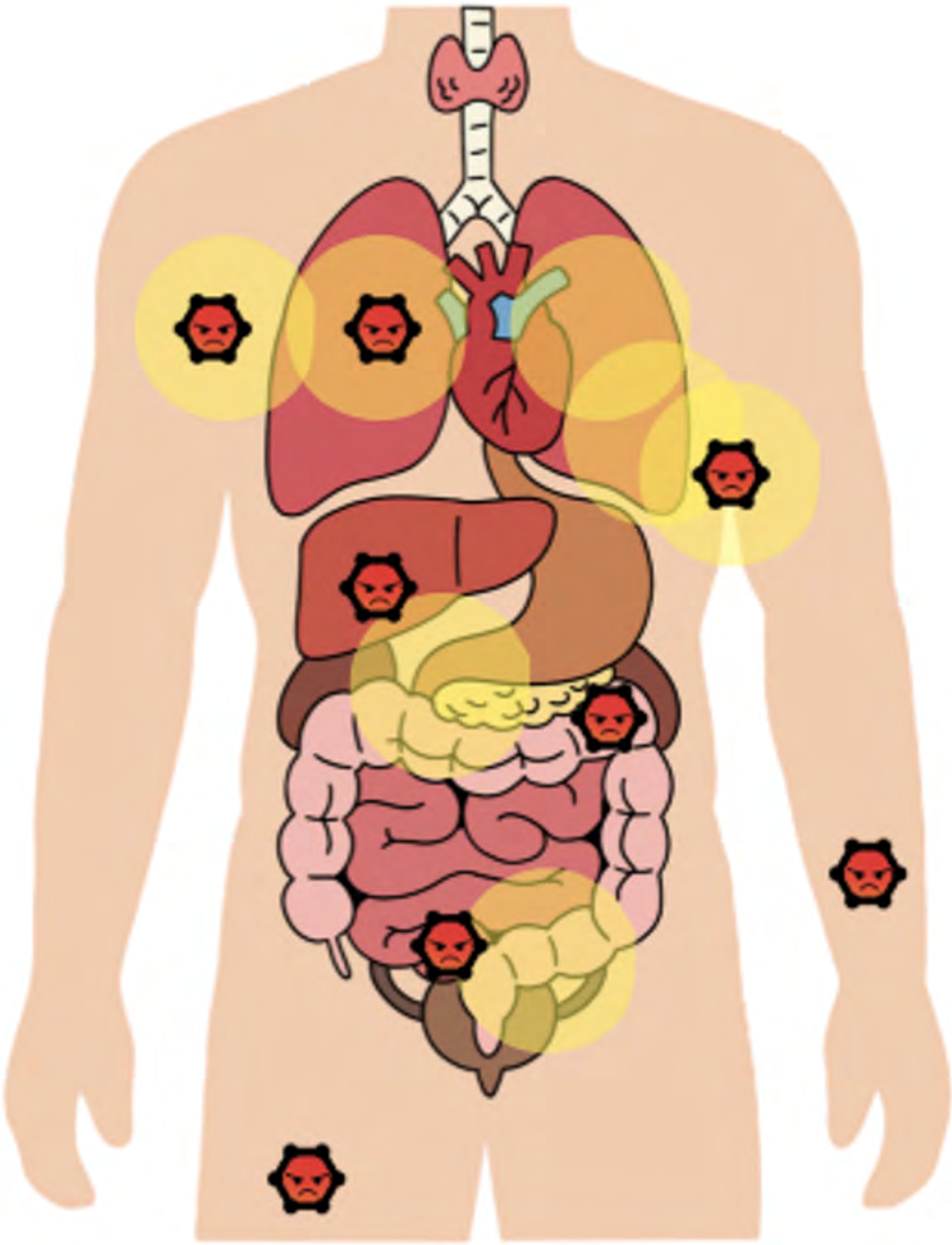
- Monomer: ethyl-, propyl-, or butyl-methacrylate
- Plasticizer: phthalates (i.e. dibutyl phthalate), citrates (i.e. tributyl acetylcitrate)
- Dimethacrylate cross-linking agent (ie ethylene glycol dimethacrylate)



Acrylic Liners: Overview



Acrylic Liners: Overview



Acrylic Liners: COE-SOFT



Table 1

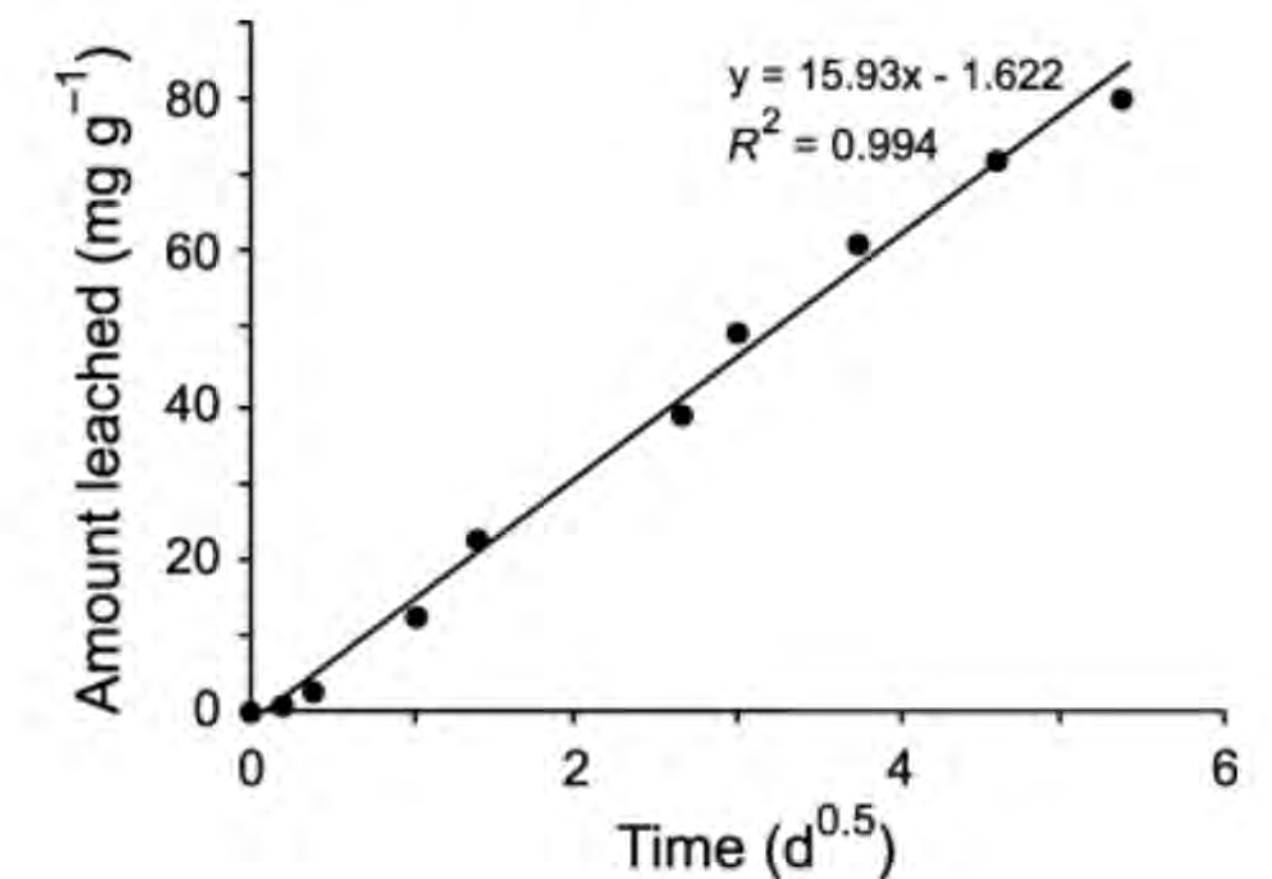
Leaching of plasticizers and hydrolysis products from Coe-Soft immersed in 0.01 M phosphate containing 0.1% Triton_{red}, pH 6.5, with or without 50 mU ml⁻¹ of esterase

Time, d	Without esterase			With esterase				
	DBP	B-SAL	M-SAL	DBP	B-SAL	M-SAL	MBP	SAL
1	70.1 (11.1)	56.0 (8.9)	17.3 (1.01)	5.3 (2.3)	47.6 (10.4)	16.3 (0.7)	84.9 (18.4)	20.8 (0.7)
7	215.2 (13.2)	168.7 (9.8)	24.8 (1.5)	27.6 (11.3)	133.8 (11.7)	21.8 (0.9)	306.7 (146)	78.8 (6.7)
14	340.8 (17.6)	268.8 (11.6)	25.8 (1.8)	27.8 (11.3)	176.2 (14.6)	21.8 (0.9)	394.2 (19.3)	161.8 (7.7)
28	441.6 (25.6)	355.6 (14.0)	25.8 (1.8)	27.8 (11.3)	194.6 (15.9)	21.8 (0.9)	450.7 (24.9)	261.1 (7.4)

B-SAL, benzylsalicylate; DBP, dibutylphthalate; MBP, monobutylphthalate; M-SAL, methylsalicylate; SAL, salicylic acid; Triton_{red}, polyoxyethylene(10)isooctylcyclohexyl ether.

Data are expressed as $\mu\text{mol g}^{-1}$ (standard deviation).

Leaching of plasticizers from temporary denture soft lining materials



Acrylic Liners: COE-SOFT



2019 COE-SOFT MSDS

3 Composition/information on ingredients

- **Chemical characterization: Mixtures**
- **Description:** Mixture of the substances listed below with nonhazardous additions.

- **Dangerous components:**

CAS: 118-58-1	benzyl salicylate	25 - < 50%
CAS: 64-17-5	ethyl alcohol	10 - < 25%
CAS: 8006-90-4	peppermint oil	0.5 - < 1%

- **Additional information:**

If a substance is marked with **, then substance is a trade secret. This is allowed under OSHA's Hazard Communication Standard (HCS) as a trade secret and under GHS as Confidential Business Information (CBI).



NO DBP LISTED





Shiver, Lauren

To: sds@gcamerica.com; SDS.gcamerica@gc.dental

Cc: ☒ Servine, Sarah; mark.heiss@gc.dental

Good afternoon!

My name is Lauren Shiver, and I am a third-year dental student at the Dental College of Georgia. A few of my classmates and I are creating a presentation on soft denture liners. I am reaching out with a few questions regarding the contents of COE-SOFT. I am particularly looking into the leaching of plasticizers from acrylic denture liners.

I have been reviewing the literature, and I came across an older (2005) article reporting that your product COE-SOFT contained certain phthalate plasticizers (for example dibutyl phthalate, listed among the ingredients. I also found some states

Has the formulation of COE-SOFT changed since 2005 (to remove salicylate or other plasticizers)? Is the current formulation provide (or direct me to) documentation or a statement whether these are compliant with any relevant phthalate regulations?

Thank you very much for your time and help. I look forward to your response.

Best,

Lauren Shiver
Dental College of Georgia
DMD Candidate | Class of 2027
(706)-833-1214
lshiver@augusta.edu



Reply



Reply all



Forward



44 PM

Thu 11/13/2025 6:17 PM



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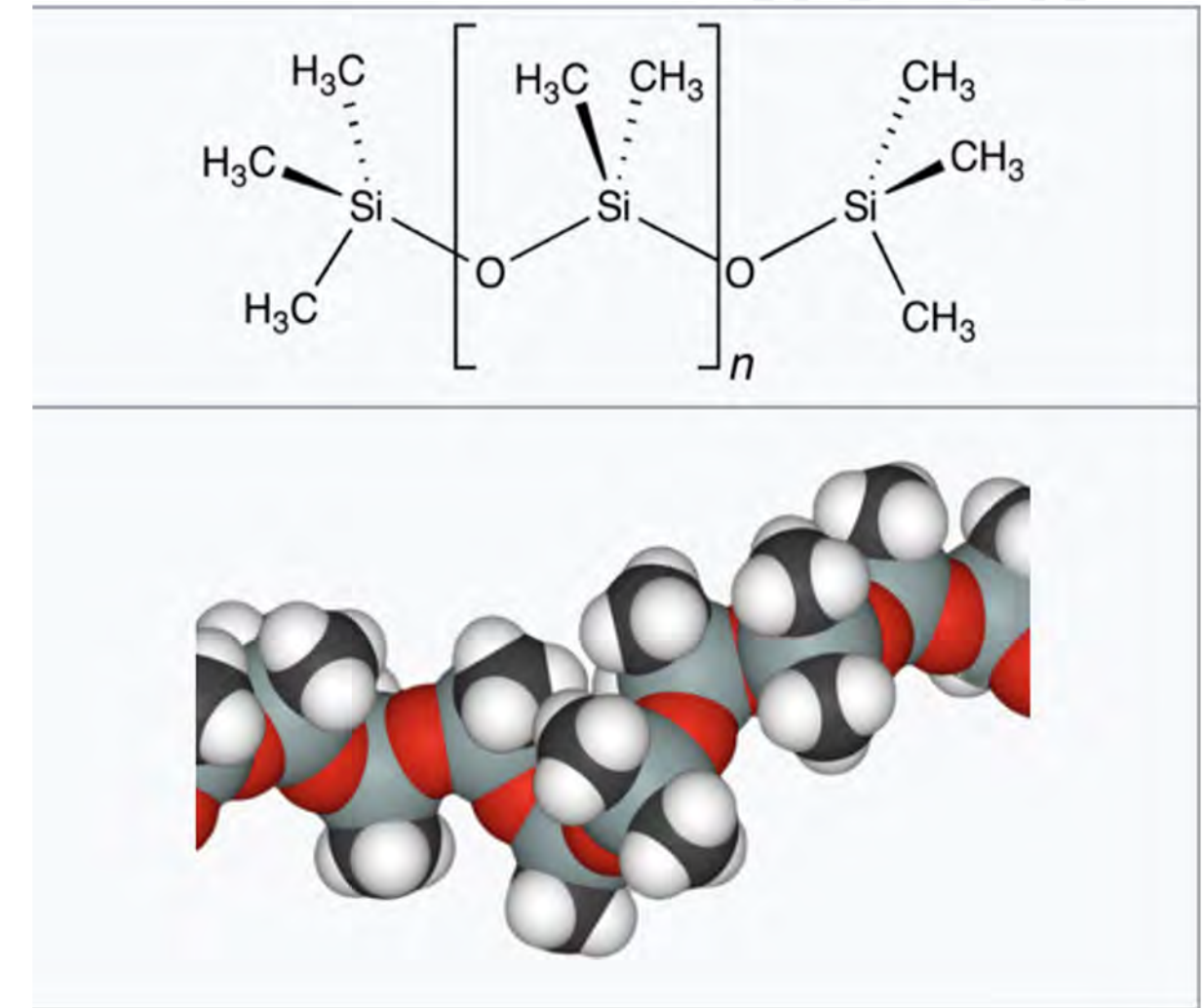


AUGUSTA UNIVERSITY
**DENTAL COLLEGE
OF GEORGIA**

Silicone Liners: Overview

- **Auto-polymerized**

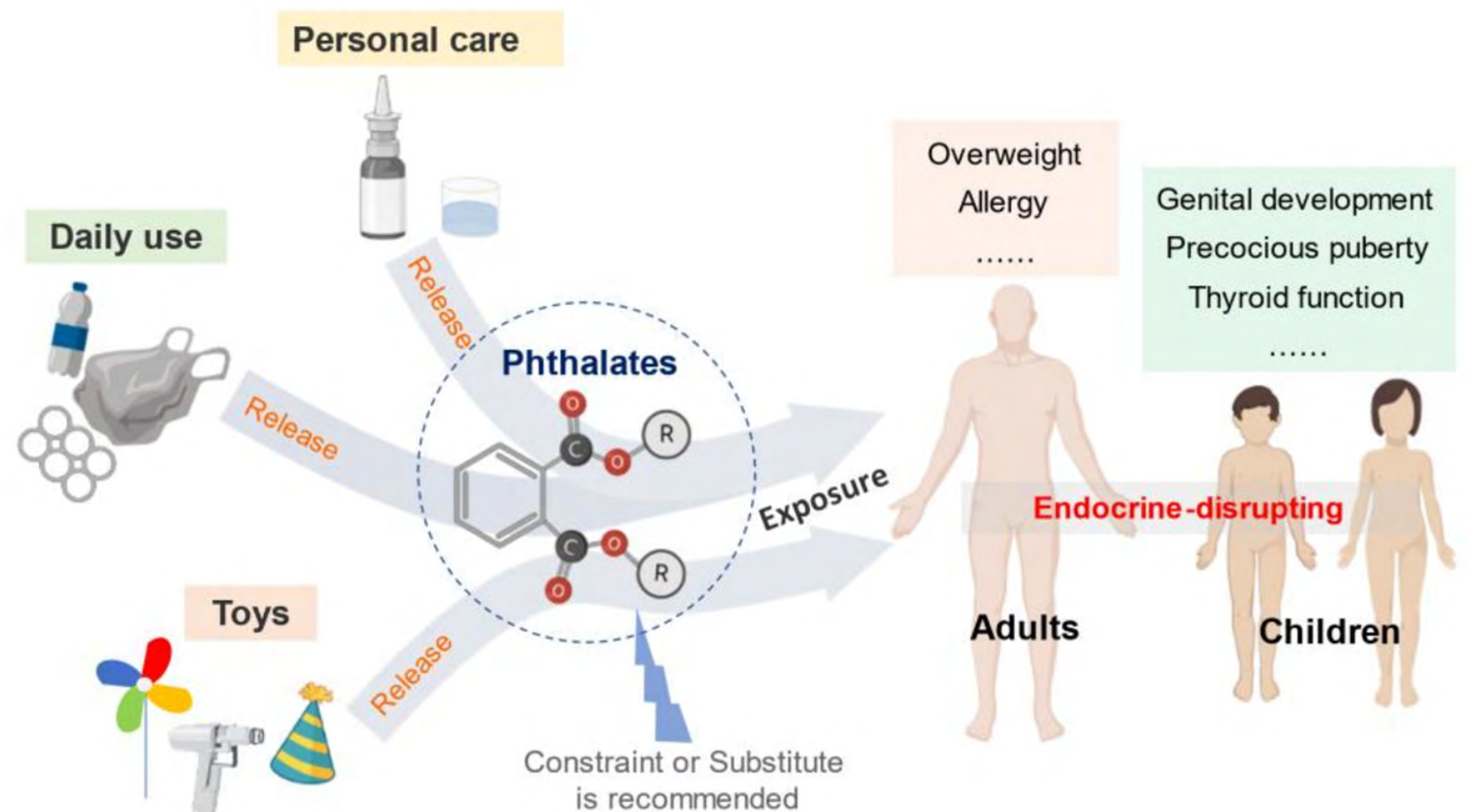
- Two paste system that cross links at room temp
 - Catalyst paste: vinyl-terminated polydimethylsiloxanes with a platinum catalyst
 - Base paste: vinyl-terminated polydimethylsiloxanes with hydride-terminated polydimethylsiloxanes
 - Mix pastes → hydrosilylation rxn → addition of Si–H from the hydride functional siloxanes results in bonds across the unsaturated bonds, forming vinyl functional siloxanes
- Low temperature required for cross-linking → can be cured in the mouth (“chair-side”)



Phthalates and Their Health Effects



- Can leach out from denture liners over time
- Myriad of phthalates with different properties & toxicities
- Difficult to generalize health effects
- Some consistent effects identified in research



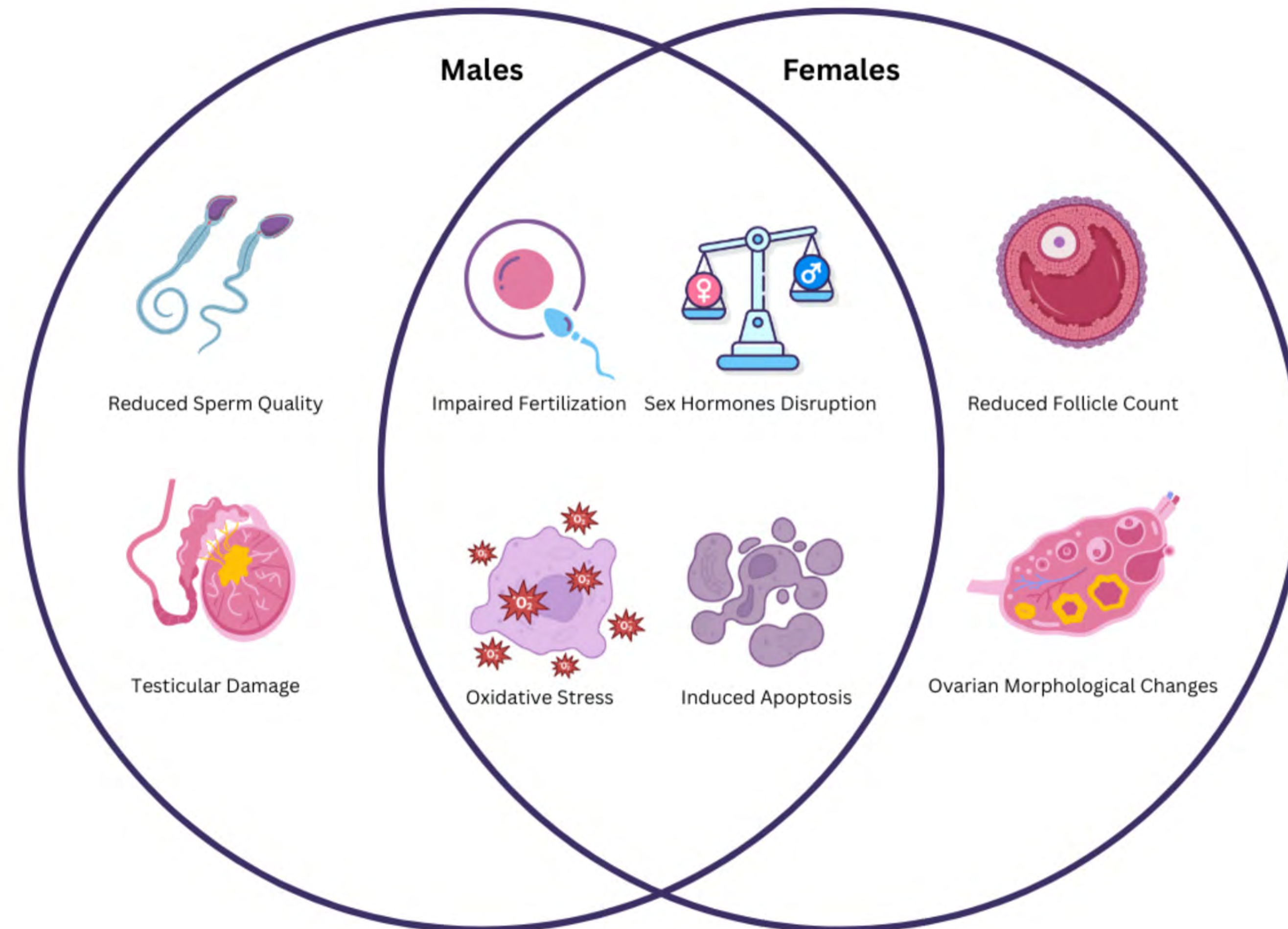
Hormone Disruption



- **Most consistent and well supported mechanistic effect**
- Endocrine disruptors → interfere with hormone signalling
- Thyroid homeostasis → disrupt T3, T4 levels
- Reduce aromatase → lower production of estrogen
- Anti-androgenic → lower production of testosterone
- Reproductive effects
 - Puberty, menstruation, fertility



Reproductive Toxicity



Cytotoxicity



- Increased reactive oxygen species
- Mitochondrial damage
- DNA strand breaks
- Disruption of membrane integrity
- Disruption of cell cycle regulation
- Found in Sertoli cells, Leydig cells, granulosa cells, macrophages, thyroid follicular cells, keratinocytes, fibroblasts, etc.

FDA Approval



- **Class II dental devices** → require safety and effectiveness data
- No liner that we have found has full FDA approval (class III)
 - Very few have publicly available 510(k) documents
 - Not full FDA approval, simply means it meets basic safety standards
 - COE-SOFT has a 510(k)
- No guaranteed long term biocompatibility or absence of health risks
 - **Plasticizer leaching and monomer release are not discussed in these documents**



BO4. Compare and contrast the effectiveness and limitations of soft liners versus alternative impression materials and techniques based on current scientific evidence and product availability.



Search Strategy



P: Edentulous patients needing accurate impressions

I: Soft liners as impressions

C: Alternative techniques

O: Accuracy/Patient Comfort/Durability

Databases used

Pubmed

Google Scholar

Keywords

Soft liners, functional impressions, SDL, denture, complete dentures

Pubmed search string

((Soft liners) OR (SDL)) AND ((Functional impression) OR (Impression)) AND ((Denture) OR (Complete Denture))

Search results

29 results

2 used for research

Other search strings used for supporting evidence

Highest level of evidence

Systematic Review and Systematic Review of RCTs



SDL Functional Relines - Advantages



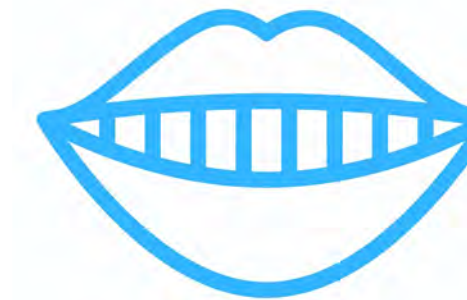
Improves soft tissue compatibility



Useful for chronic ill-fitting dentures in order to capture full range of motion



Adjunctive tx for denture stomatitis



Increased comfort of immediate dentures, while also capturing functional movements



No distortion when removed from the mouth - SDL in “plastic stage”



SDL Functional Relines - Disadvantages



Existing denture required

- normally remove dentures ~48 hrs before final impressions
- Some patients are against this



Increased roughness of master cast



Insufficient research for how long it should be worn



Increased plaque retention and susceptibility for candida



Must be poured IMMEDIATELY due to viscoelasticity



Material choice?



2019 Systematic Review answering the question, “Which final impression technique and material is best for complete and removable partial dentures”

“There is **no clear evidence** that one technique or material has a substantial advantage over another for making complete dentures and removable partial dentures. Available evidence for the relative benefits of different denture fabrication techniques and final-impression materials is limited and is of low or very low quality.”

Material choice cont'd



❖ Coe-Soft Reline

➤ \$200 kit with roughly 32 uses = \$6.25/use →

\$12.50/patient

❖ PVS

➤ Flexitime Heavy Body

■ \$24.50/cartridge

➤ Flexitime Light Body

■ \$18/cartridge

❖ Primobase - Light Cured Denture Base Material

➤ \$2/piece

❖ Shining 3D AOralscan 3

➤ \$12,999

\$46.50/patient





**“An ideal impression must be
in the mind of the dentist before
it is in his (or her) hands”**

M.M. Devan (DDS)
Journal of Prosthetic Dentistry
1952



BO5. Apply a critical decision tree to select appropriate functional impression approaches for edentulous patients, balancing clinical outcomes, material properties, safety concerns, and regulatory considerations.



Search Strategy



PICO: Not applicable

Databases used

Pubmed

Keywords

soft liner, functional impression

Pubmed search string

((soft liner) AND (functional impression))

Search results

19 search results, 2 were used

Highest level of evidence

Systematic reviews, case study



Critical Decision tree → Functional Impression Approaches for Edentulous Patients



Assess Patient's Ridge and Mucosal Condition

Healthy, firm mucosa with adequate ridge support

Unhealthy, inflamed tissues on existing denture

Due to ill-fitting denture?

Atrophic Ridge

Flabby Ridge

Normal Ridge

Existing denture?

Functional Reline
COE-SOFT ?
Perfect fit ?
Hydrocast and flowcast ?
cheaper way

Neutral zone technique
Selective Pressure Approach



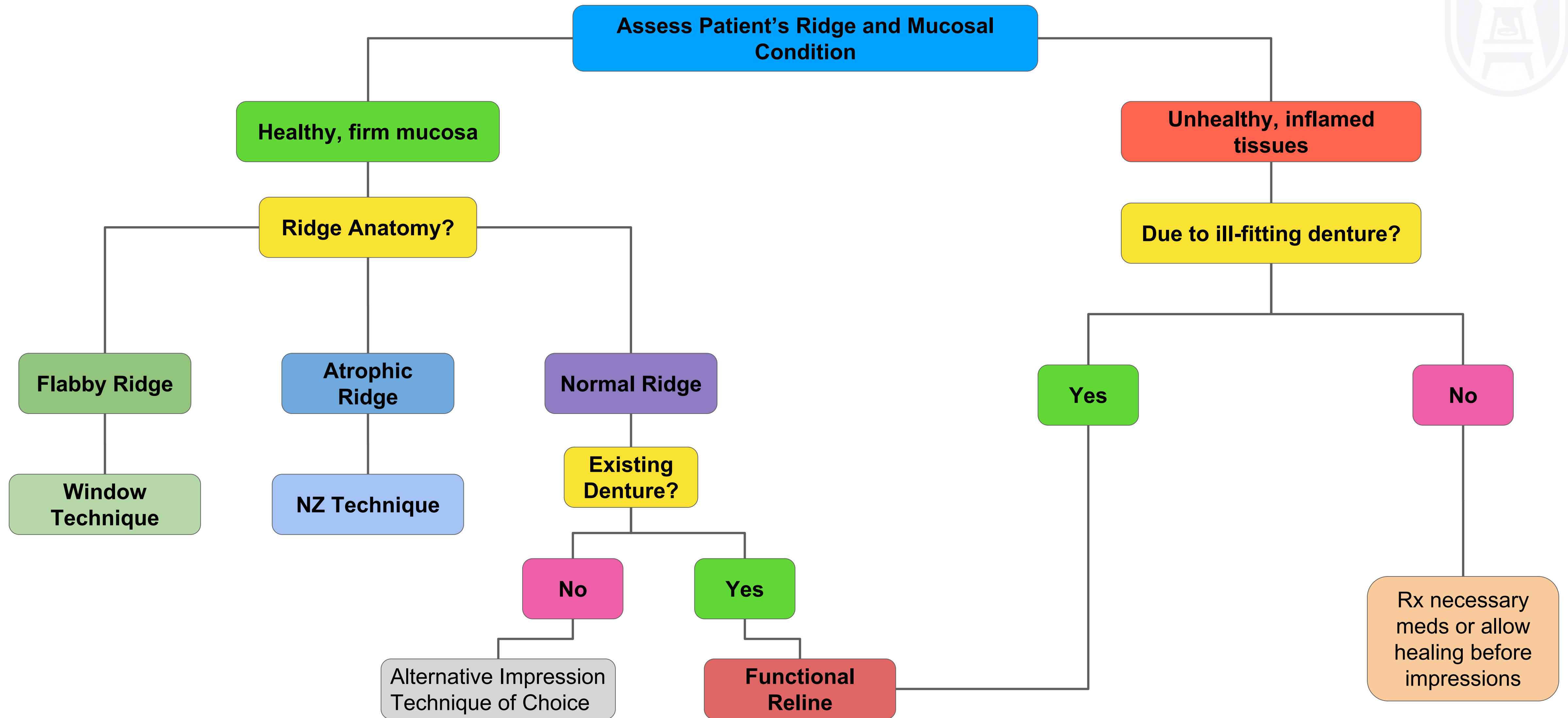
Window Technique
Mucostatic and Selective Pressure Approach



Custom Tray
Cost more money, longer time



Critical Decision tree → Functional Impression Approaches for Edentulous Patients



Summary of BOs



BO1 – Biological & Clinical Principles of Functional Impression Techniques

- Capture oral tissues functionally for accurate denture fit and comfort.
- Core principles: retention, stability, support, esthetics, and tissue preservation.
- Ideal materials show viscoelasticity, flow, dimensional stability, and are biocompatible.
- Impression theories: mucostatic (at rest), mucocompressive (under pressure), selective pressure (controlled stress).
- Common techniques: functional reline, fluid wax, closed/open mouth, window (flabby ridge), neutral zone (atrophic ridge).
- Soft liners act as functional impression materials by dynamically adapting to tissues and enhancing comfort.

Summary of BOs



BO2 – Differentiate Between Soft Liner Products

- **Types:**
 - **Acrylic-based:** polymer + monomer/plasticizer (e.g., Coe-Soft); soft initially, harden over time.
 - **Silicone-based:** polydimethylsiloxane (e.g., Molloplast-B); stable, longer-lasting, colorfast.
- **Classification:**
 - *Short-term (tissue conditioners):* for healing or immediate post-surgery (days–1 month).
 - *Long-term:* acrylic (up to 6 months) or silicone (up to 1 year).
- **Uses:** Cushion between denture and mucosa, redistribute pressure, relieve soreness.
- **Off-label use:** Employed for functional impressions to reline and re-adapt existing dentures.
- **Performance:** Silicone > acrylic in water resistance, color stability, and bond strength.

Summary of BOs



BO3 – Evaluate Health Risks & Patient Safety

- Phthalates/plasticizers used for flexibility can leach and **act as endocrine disruptors**.
- **Material-specific:**
 - Acrylic: risk of plasticizer/monomer leaching
 - i. Phthalate release → hormonal disruption, microbial colonization (Candida), surface roughness
 - Silicone: fewer chemical risks, higher microbial potential
 - i. Risks with the adhesives
- **Regulatory:** FDA Class II devices. Limited long-term safety data.
- Examples: *Coe-Soft* (phthalate-free); *Flowcast* (discontinued in EU for excessive phthalates).

Summary of BOs



BO4 – Compare Soft Liners vs. Alternative Impression Materials

- **Advantages:**
 - Conform to dynamic tissues and improve comfort.
 - Enhance retention and reduce sore spots.
 - Useful for ill-fitting or resorbed ridges.
- **Limitations:**
 - **Need existing denture**, promote plaque/fungal growth.
 - Roughen dental stone surfaces, must pour immediately.
- **Alternatives:** fluid wax, zinc eugenol, elastomers – more precise but less cushioning.
- **Evidence:** No single superior material/technique; clinical success depends on patient anatomy and clinician expertise.
- Soft liners serve as an **adjunct**, not a replacement for traditional methods.



Summary of BOs



BO5 – Apply a Critical Decision Tree for Impression Selection

- **Assess Patient Tissue Condition:**
 - Healthy ridge → mucostatic/selective pressure.
 - Flabby ridge → window technique.
 - Atrophic ridge → neutral zone technique.
 - Inflamed tissues → functional reline/tissue conditioner.
- **Select Material:**
 - Acrylic (short-term) for healing or interim comfort.
 - Silicone (long-term) for durability and maintenance.
- **Clinical & Regulatory Considerations:**
 - Verify biocompatibility and FDA status.
 - Balance comfort, hygiene, cost, and longevity.
- **Outcome Goal:** Achieve optimal function, safety, and patient satisfaction while ensuring compliance with standards.



Sources



BO1:

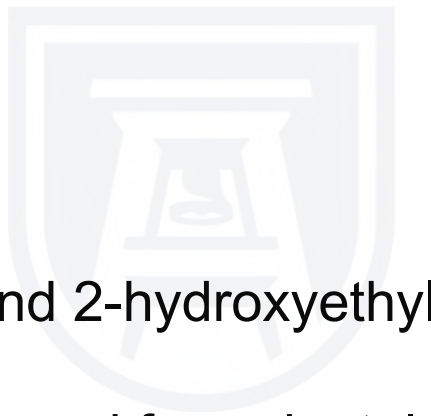
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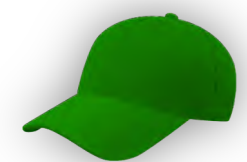
Red Hat (Patient): Expresses emotions, concerns, and preferences regarding the material.



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